

Dual Converters

Anti-parallel arrangement of two single-phase or two three-phase fully controlled FWRs

Different Modes Of Operation of Dual converter

- Non-circulating current (circulating current free) mode of operation.
- Circulating current mode of operation.

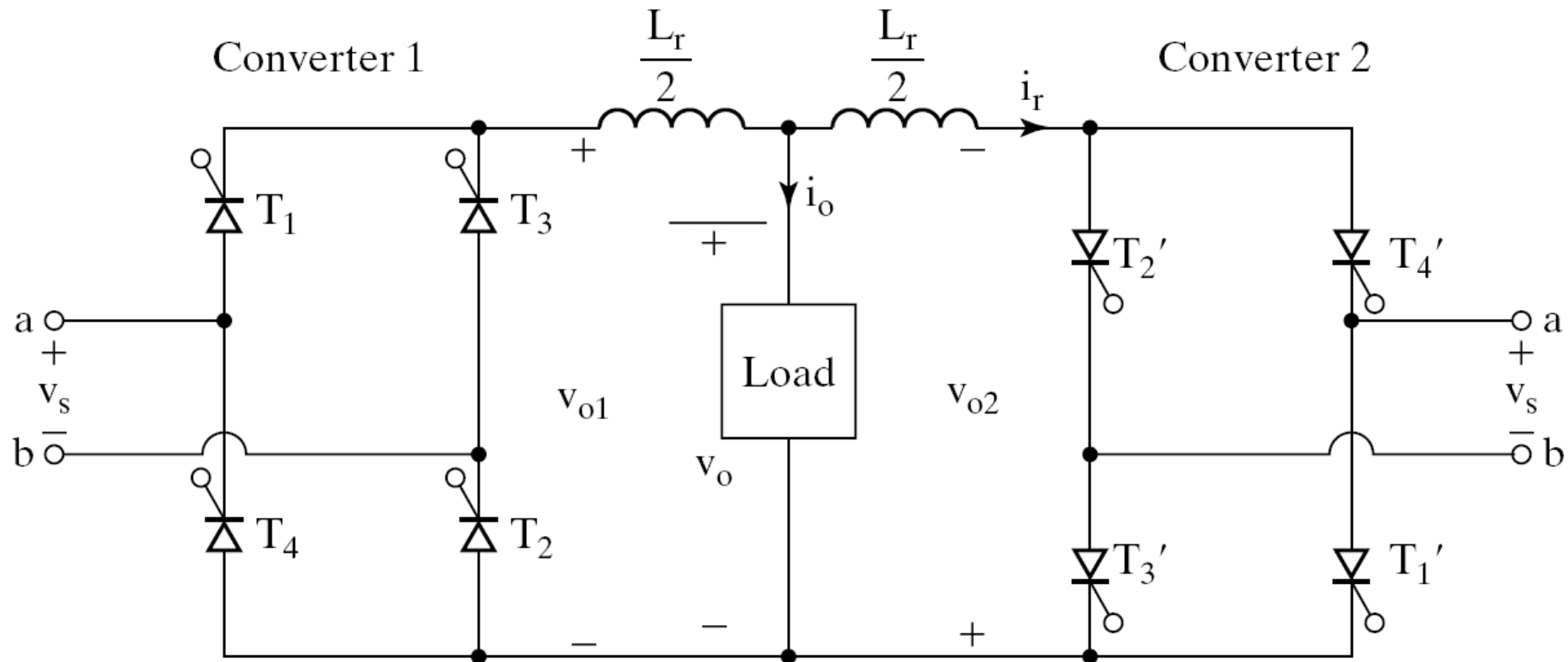
Non-Circulating Mode of Operation

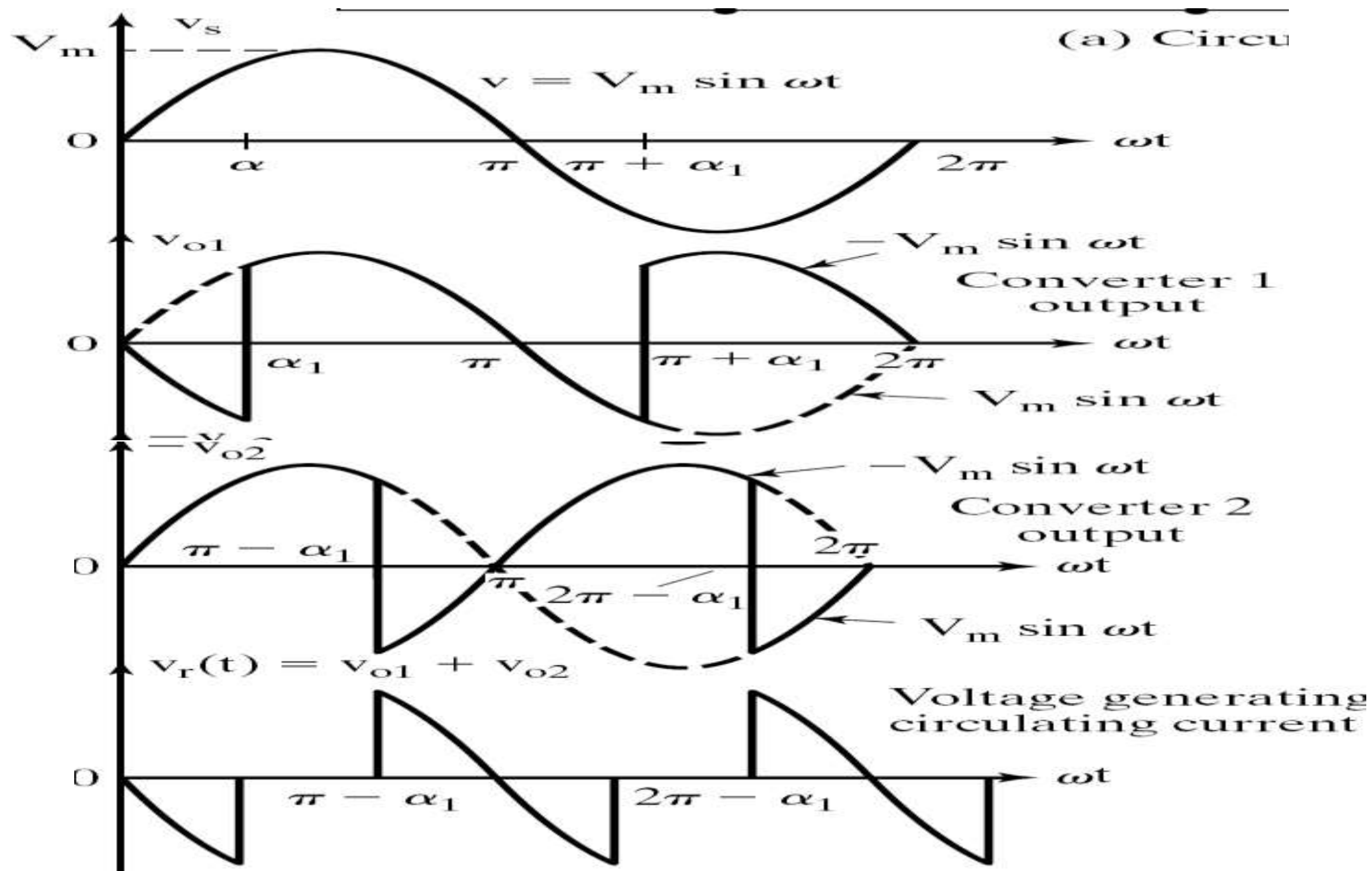
- In this mode only one converter is operated at a time. Circulating current reactor is not needed.
- When converter 1 is ON, $0 < \alpha_1 < 90^\circ$
- V_{dc} is positive and I_{dc} is positive.
- When converter 2 is ON, $0 < \alpha_2 < 90^\circ$
- V_{dc} is negative and I_{dc} is negative.

Circulating Current Mode Of Operation

- In this mode, both the converters are switched ON and operated at the same time.
- The trigger angles α_1 and α_2 are adjusted such that $(\alpha_1 + \alpha_2) = 180^\circ$; $\alpha_2 = (180^\circ - \alpha_1)$.

Single-phase Dual Converter





The average dc output voltage of converter 1 is

$$V_{dc1} = \frac{2V_m}{\pi} \cos \alpha_1$$

The average dc output voltage of converter 2 is

$$V_{dc2} = \frac{2V_m}{\pi} \cos \alpha_2$$

In the dual converter operation one converter is operated as a controlled rectifier with $\alpha < 90^\circ$ & the second converter is operated as a line commutated inverter in the inversion mode with $\alpha > 90^\circ$

$$\therefore V_{dc1} = -V_{dc2}$$

$$\frac{2V_m}{\pi} \cos \alpha_1 = \frac{-2V_m}{\pi} \cos \alpha_2 = \frac{2V_m}{\pi} (-\cos \alpha_2)$$

$$\therefore \cos \alpha_1 = -\cos \alpha_2$$

or

$$\cos \alpha_2 = -\cos \alpha_1 = \cos (\pi - \alpha_1)$$

$$\therefore \alpha_2 = (\pi - \alpha_1) \text{ or}$$

$$(\alpha_1 + \alpha_2) = \pi \text{ radians}$$

Which gives

$$\alpha_2 = (\pi - \alpha_1)$$

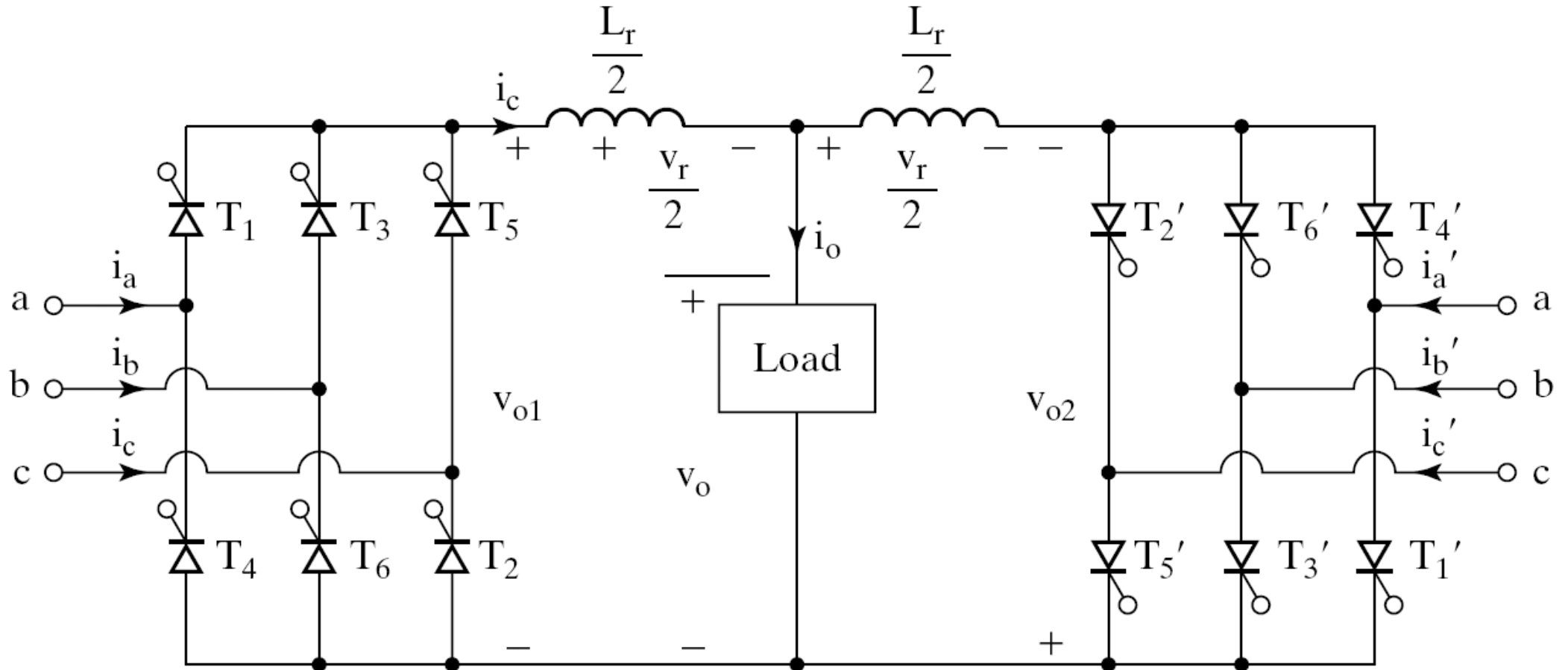
Advantages of Circulating Current Mode Of Operation

- The circulating current maintains continuous conduction of both the converters over the complete control range, independent of the load.
- One converter always operates as a rectifier and the other converter operates as an inverter; the power flow in either direction at any time is possible.
- As both the converters are in continuous conduction, it results in faster dynamic response. i.e., the time response for changing from one quadrant operation to another is faster.

Disadvantages of Circulating Current Mode of Operation

- There is always a circulating current flowing between the converters.
- When the load current falls to zero, there will be a circulating current flowing between the converters. Hence, circulating current reactors have to be employed in order to limit the peak circulating current to a safe level.
- The converter thyristors should be rated to carry a peak current much greater than the peak load current.

Three-phase Dual Converter



Four Quadrant Operation

Conv. 2
Inverting
 $\alpha_2 > 90^\circ$

Conv. 1
Rectifying
 $\alpha_1 < 90^\circ$

Conv. 2
Rectifying
 $\alpha_2 < 90^\circ$

Conv. 1
Inverting
 $\alpha_1 > 90^\circ$

